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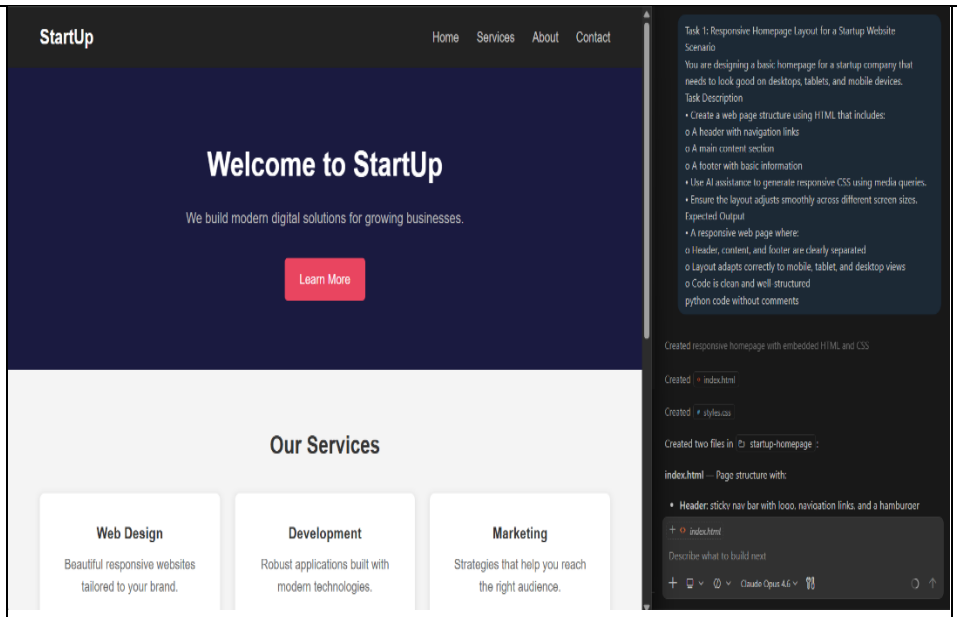
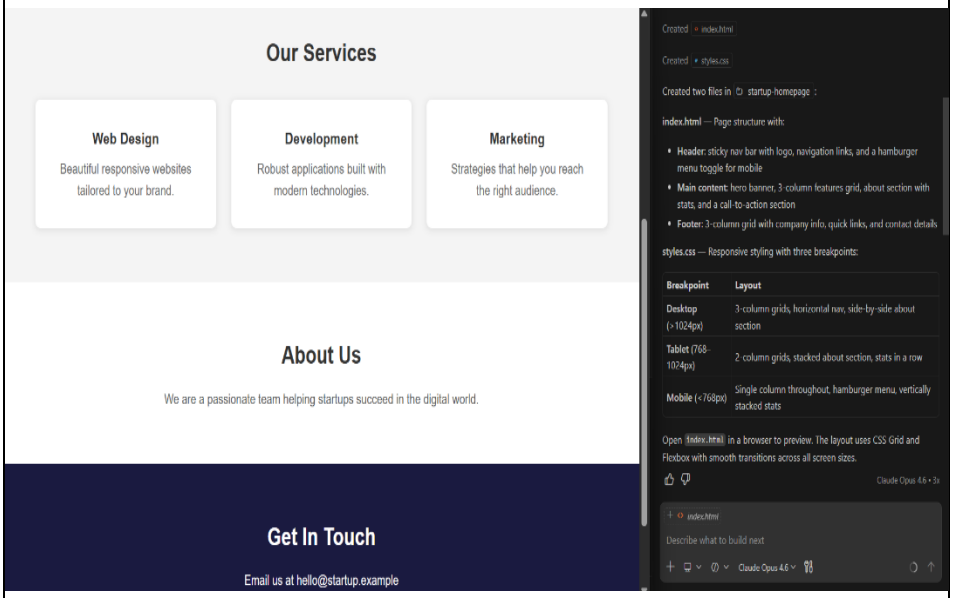
SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name:B. Tech		**Assignment Type: Lab**	**Academic Year:**2025-2026
Course Coordinator Name	Dr. Rishabh Mittal		
Instructor(s)Name	Mr. S Naresh Kumar Ms. B. Swathi Dr. Sasanko Shekhar Gantayat Mr. Md Sallauddin Dr. Mathivanan Mr. Y Srikanth Ms. N Shilpa Dr. Rishabh Mittal (Coordinator) Dr. R. Prashant Kumar Mr. Ankushavali MD Mr. B Viswanath Ms. Sujitha Reddy Ms. A. Anitha Ms. M.Madhuri Ms. Katherashala Swetha Ms. Velpula sumalatha Mr. Bingi Raju Mr. G. Kranthi		
Course Code	23CS002PC304	**Course Title**	AI Assisted Coding
Year/Sem	III/I	**Regulation**	R23
Date and Day of Assignment	Week 7 - Thursday	**Time(s)**	23CSBTB01 To 23CSBTB52
Duration	2 Hours	**Applicable to Batches**	All Batches
AssignmentNumber:14.4 (Present assignment number)/**24**(Total number of assignments)			

Q.No.	Question	*ExpectedTime to complete*

1	**Lab 14: Web Frontend Development – AI-Assisted HTML/CSS/JS Generation** **Lab Objectives** • Design **functional, visually appealing** web applications using HTML, CSS, and JavaScript with AI assistance. • Apply **responsive, accessible, and interactive design principles**. • Create **practical UI components** for real-world web applications. • Use AI to optimize **layout, UX, and performance**. **Lab Outcomes** • Applying AI assistance to frontend development tasks • Understanding how HTML, CSS, and JavaScript work together • Improving responsiveness, interactivity, and usability • Evaluating and refining AI-generated frontend code	Week 7

	Task 1: Responsive Homepage Layout for a Startup Website
	
	Scenario
	
	You are designing a **basic homepage for a startup company** that needs to look good on desktops, tablets, and mobile devices.
	
	Task Description
	
	• Create a web page structure using HTML that includes:
	  ○ A header with navigation links
	  ○ A main content section
	  ○ A footer with basic information
	• Use AI assistance to generate responsive CSS using media queries.
	• Ensure the layout adjusts smoothly across different screen sizes.
	
	Expected Output
	
	• A responsive web page where:
	  ○ Header, content, and footer are clearly separated
	  ○ Layout adapts correctly to mobile, tablet, and desktop views
	  ○ Code is clean and well-structured	





Task 2: Interactive Call-to-Action Button

Scenario

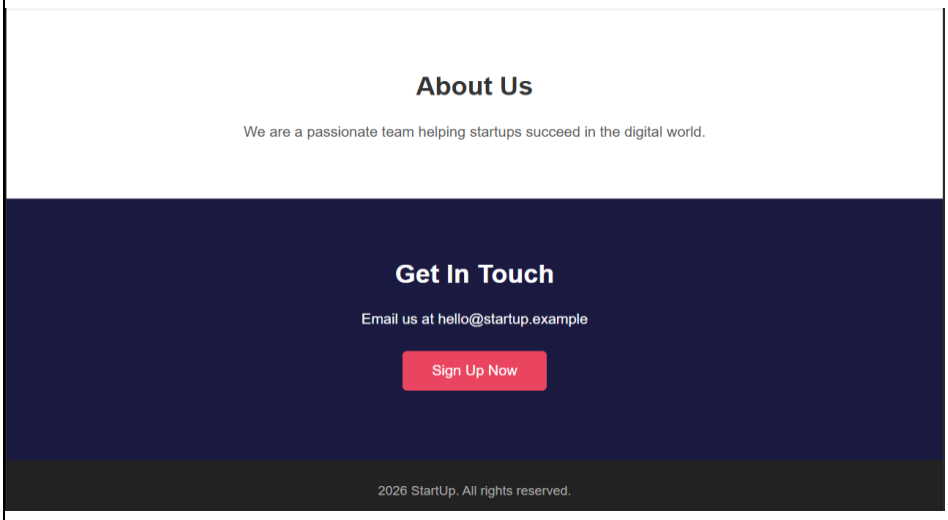
A marketing page requires a **call-to-action button** that responds when users interact with it.

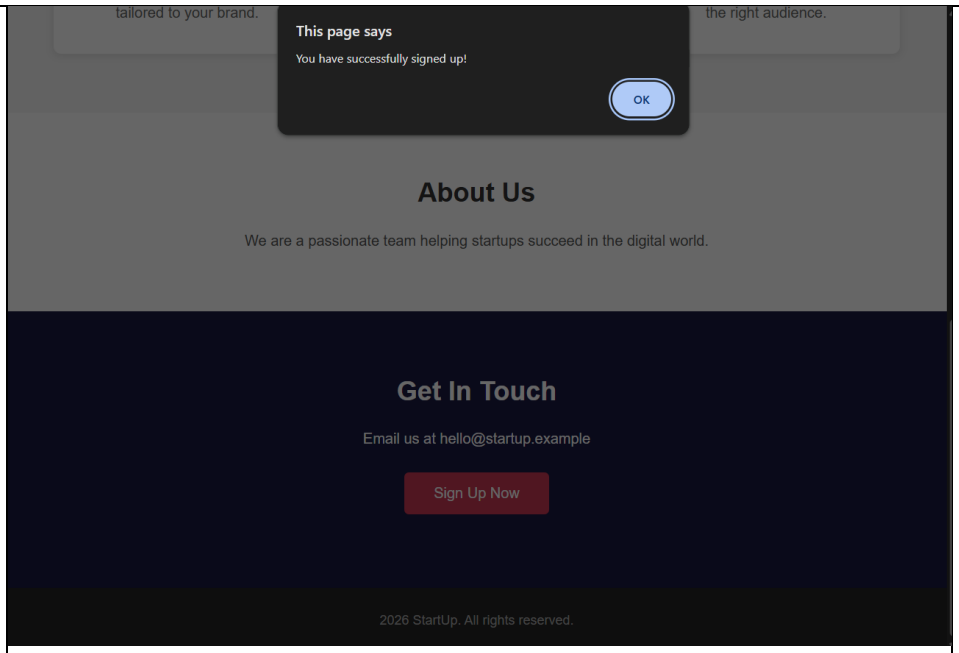
Task Description

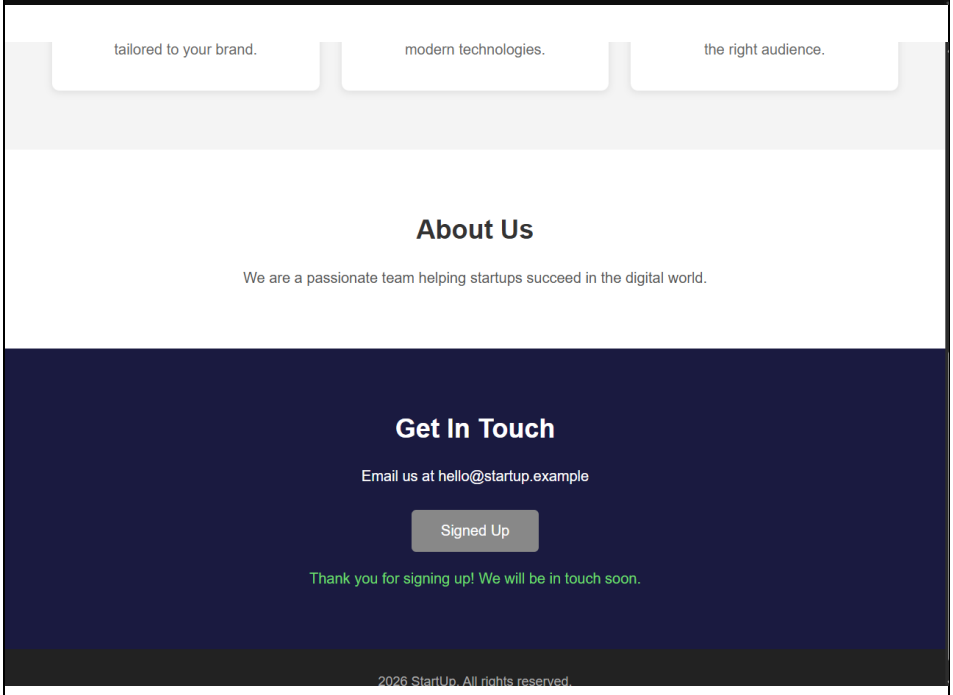
- Add a button to a web page using HTML.
- Use AI to generate JavaScript that responds to the button click by:
  - Displaying a message or alert
- Ask AI to add meaningful comments explaining the JavaScript logic.

Expected Output

- A web page where:
  - Clicking the button triggers a visible response (alert or message)
  - JavaScript code is readable and commented
  - No page reload occurs







Task 3: Contact Form with Client-Side Validation

Scenario

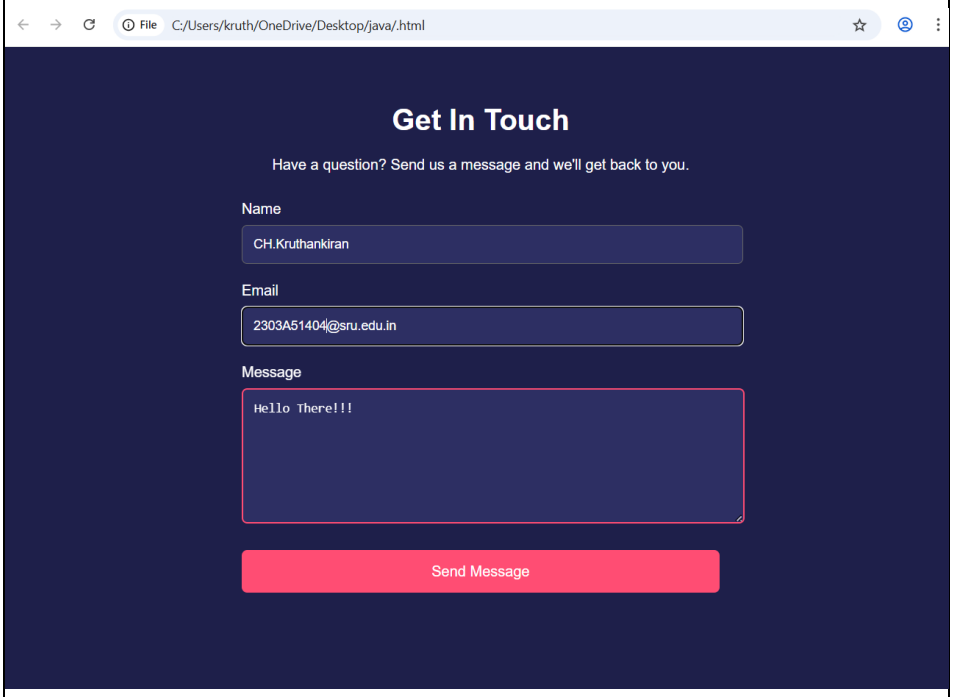
You are developing a **contact form for a company website** to collect user queries safely and correctly.

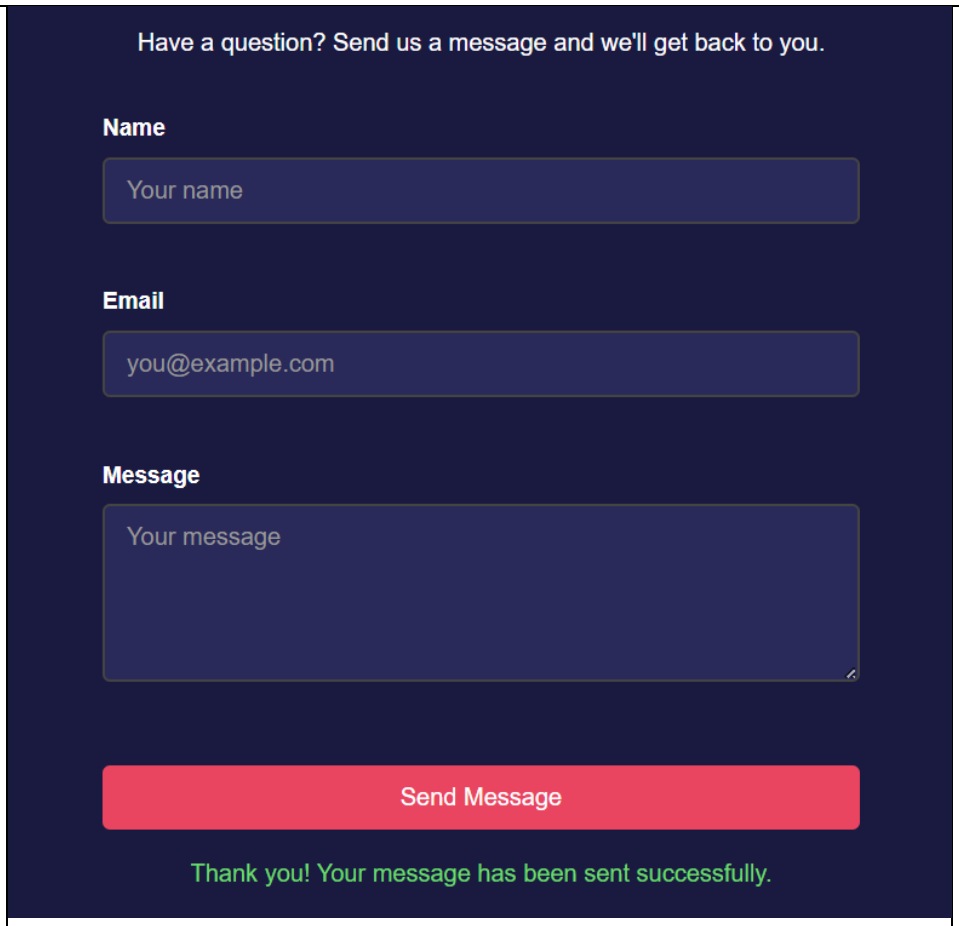
Task Description

- Design a form with fields for:
  - Name
  - Email
  - Message
- Use AI to generate JavaScript validation logic that:
  - Prevents empty submissions
  - Checks for a valid email format
- Display clear error messages for invalid input.

Expected Output

- A functional form where:
  - Invalid input shows inline error messages
  - Valid input displays a success message
  - Validation happens on the client side using JavaScript





Task 4: Dynamic List Management for a Product Page

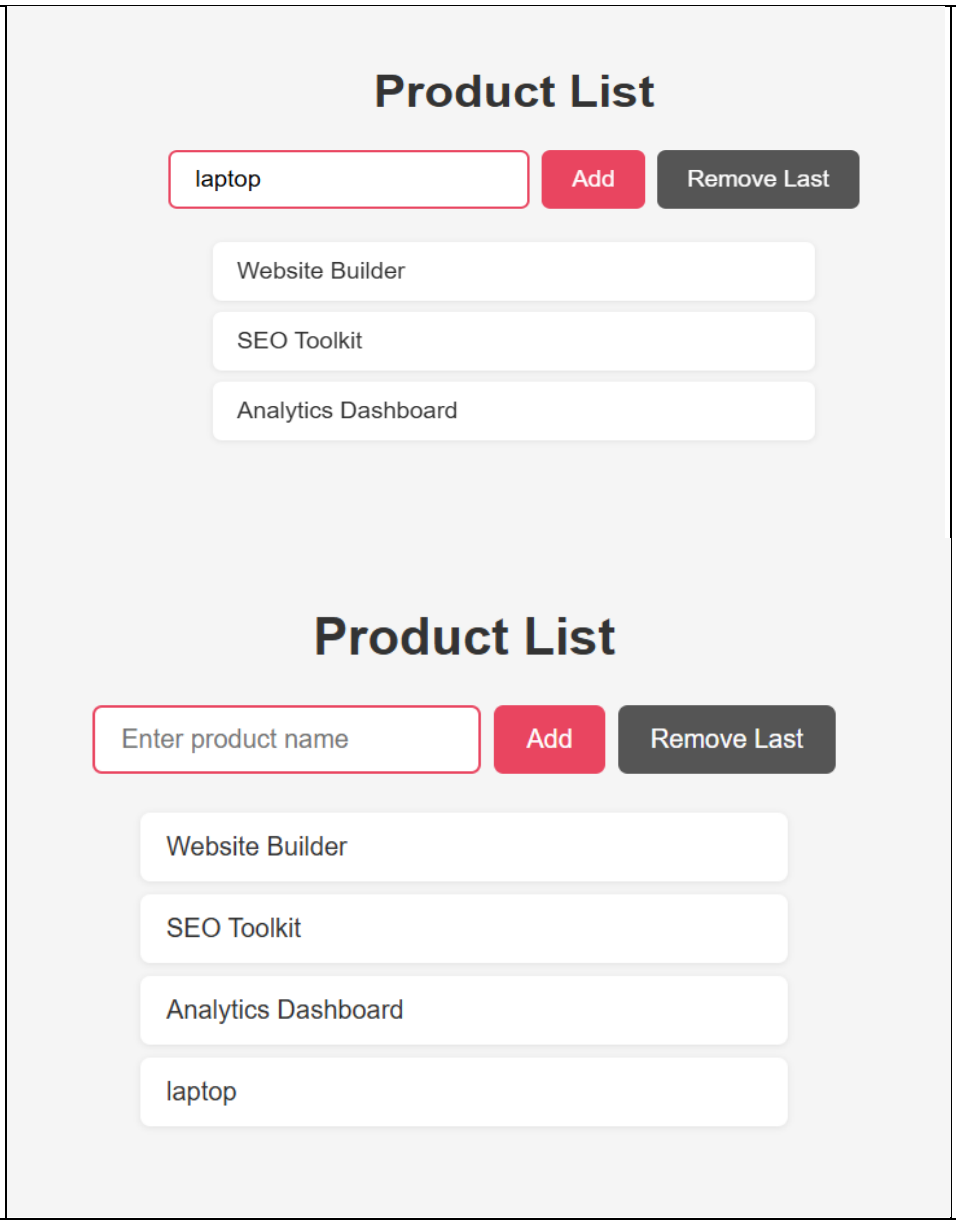
Scenario

A product listing page needs to **add or remove items dynamically** without refreshing the page.

Task Description

- Create a simple HTML list to display items.
- Use AI to generate JavaScript that:
  - Adds a new item when an “Add” button is clicked
  - Removes the last item when a “Remove” button is clicked
- Ensure the DOM updates dynamically.

	Expected Output
	
	• A web page where:
	  ○ Items can be added and removed dynamically
	  ○ Changes appear instantly without page reload
	  ○ JavaScript logic is clear and maintainable	


Product List
laptop
Add
Remove Last
Website Builder
SEO Toolkit
Analytics Dashboard
Product List
Enter product name
Add
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Website Builder
SEO Toolkit
Analytics Dashboard
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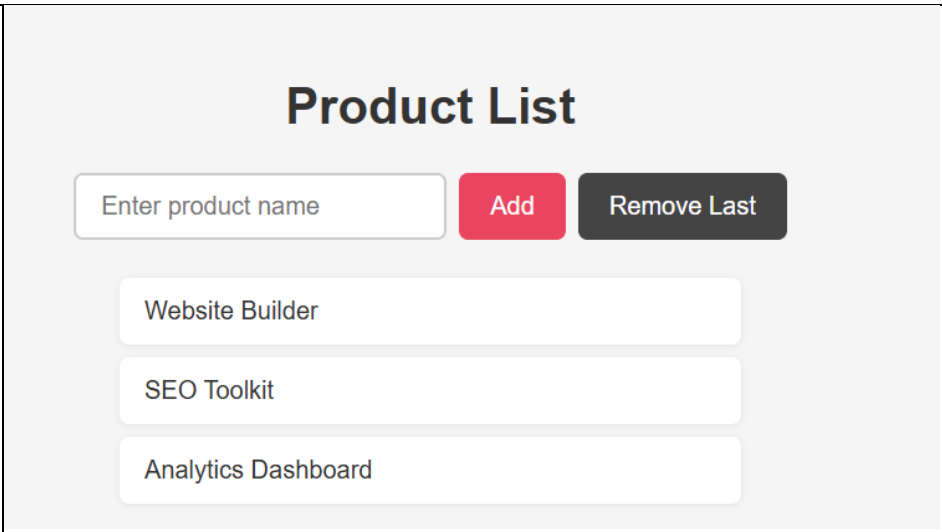


Task 5: Modal Popup for User Notifications

Scenario

You are building a **user notification popup** for announcements or alerts on a website.

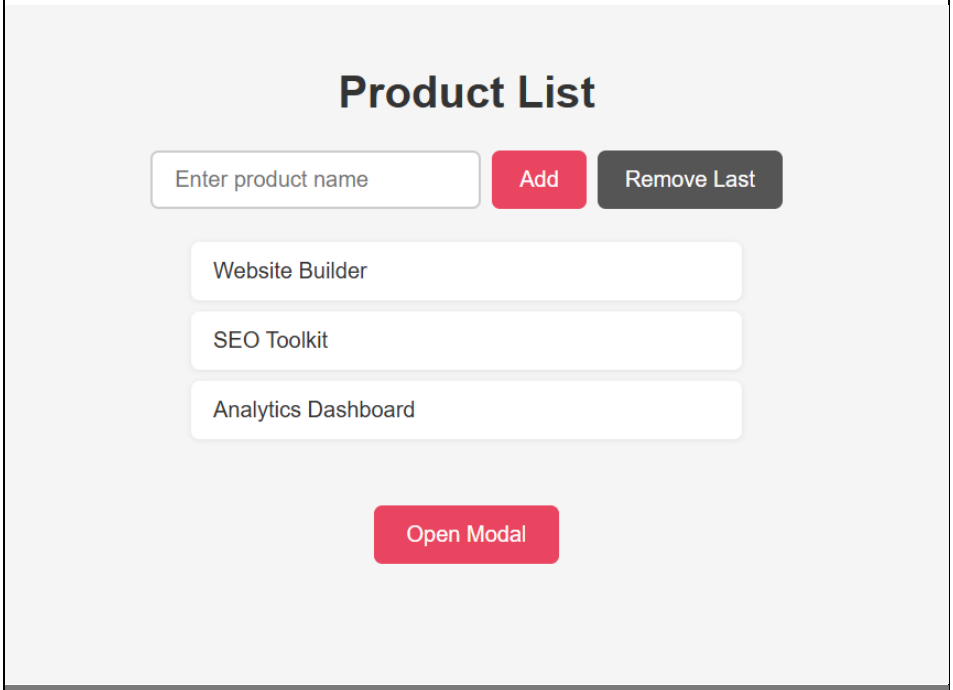
Task Description

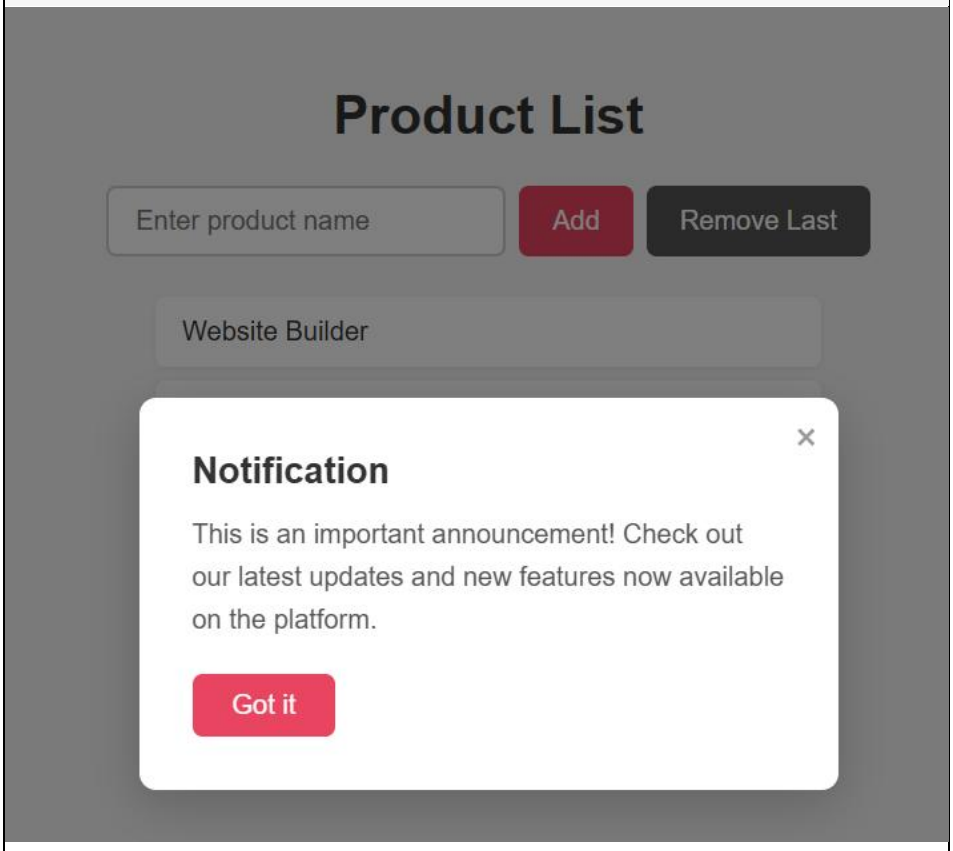
- Use AI to help generate:
  - HTML structure for a modal popup
  - CSS styling with a semi-transparent background overlay
  - JavaScript to open and close the modal
- Include multiple ways to close the modal (button or overlay click).

Expected Output

- A web page where:
  - Clicking “Open Modal” displays the popup
  - Popup overlays the page content clearly

- Modal closes smoothly when instructed





Note: Report should be submitted a word document for all tasks in a

	single document with prompts, comments & code explanation, and output and if required, screenshots	